

TATA GROUP × FORAGE

INTERNSHIP / JOB SIMULATION REPORT

GenAI Powered Data Analytics

Exploratory analytics • Delinquency prediction • Business storytelling • AI-driven collections strategy

DETAIL	INFORMATION
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PROGRAM	GenAI Powered Data Analytics Job Simulation
PLATFORM / PARTNER	Forage / Tata Insights and Quants (Tata iQ)
COMPLETION DATE	31 August 2026

Academic Submission

DECLARATION

I, ABHINAV KUMAR TRIPATHI, declare that this academic report summarizes my completion of the Tata Group GenAI Powered Data Analytics Job Simulation hosted on Forage. The report presents the work covered in the simulation, the analytical reasoning used, and the learning outcomes derived from the completed tasks. The activities were performed in a virtual job-simulation environment and are reported as such.

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ACKNOWLEDGEMENT

I sincerely thank Tata Insights and Quants (Tata iQ) and Forage for providing a structured job-simulation experience focused on practical data analytics and AI transformation. The simulation helped connect exploratory data analysis, predictive modeling, business communication, and responsible AI design into one end-to-end problem-solving workflow. I am also grateful to my academic mentors and family for their encouragement while completing this learning activity.

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1. Introduction and Simulation Overview

The Tata GenAI Powered Data Analytics Job Simulation on Forage places the learner in the role of an AI transformation consultant supporting financial-services decision making. Forage describes the experience as a self-paced, 3–4 hour simulation in which participants work through example tasks that replicate the type of work carried out by Tata iQ professionals.

The central scenario involves Geldium Finance, a financial-services company seeking to reduce credit-card delinquency through advanced analytics and AI-enabled decision making. The simulation combines technical analysis with business communication and responsible AI design.

2. Business Context and Problem Statement

The business challenge is to move from reactive collections activity toward a more proactive, data-informed approach. Customer financial and behavioral information can contain useful signals for identifying delinquency risk, but those signals must be translated into clear and responsible business actions.

The key problem is therefore not only prediction accuracy. The solution must also address data quality, interpretability, fairness, customer impact, and the point at which human review should remain in control.

3. Objectives of the Simulation

- Understand how exploratory analysis reveals patterns, missing values, and inconsistencies before modeling.
- Build a structured approach for binary delinquency prediction using customer financial and behavioral variables.
- Translate analytical findings into a specific, measurable and actionable collections recommendation.
- Design an AI-driven collections workflow that can adapt to outcomes through a learning loop.
- Apply responsible AI principles including fairness, transparency, explainability, privacy, and human oversight.

4. Methodology and Task Structure

TASK	FOCUS	CORE OUTPUT	KEY CONCEPTS
Task 1	EDA & risk profiling	Data-quality and risk observations	EDA, missing data, imputation, synthetic data
Task 2	Delinquency prediction	Structured predictive model plan	Logistic Regression, precision, recall, F1, AUC-ROC
Task 3	Business reporting	Recommendation framework	Data storytelling, SMART action, business impact
Task 4	AI-driven collections	Agentic operating model	Learning loop, fairness, human review, compliance

5. Task 1 — Exploratory Data Analysis and Risk Profiling

5.1 Purpose of EDA

Exploratory Data Analysis (EDA) is used before predictive modeling to understand the dataset's main characteristics and identify patterns, missing values, and inconsistencies. This reduces the risk of building a model on misunderstood or poor-quality data.

5.2 Handling Missing Values

The appropriate approach is to first understand why values are missing, then choose a suitable statistical imputation method such as mean, median, or regression-based imputation and validate the choice with domain knowledge. Missing values should not automatically be interpreted as zero income, zero credit history, or another substantive condition.

5.3 Synthetic Data

Synthetic data can help expand an analytical dataset while supporting privacy when real customer data is limited or sensitive. It should be validated carefully so that simulated patterns remain useful and do not introduce unrealistic relationships.

6. Task 2 — Predicting Delinquency with AI

6.1 Modeling Objective

The model is intended to estimate which customers are more likely to experience delinquency so that the Collections team can prioritize appropriate early interventions. The target is treated as a binary outcome: delinquent versus not delinquent.

6.2 Proposed Model Logic

- Inspect and clean the data; exclude Customer_ID because it is a unique identifier rather than a predictive feature.
- Perform EDA and address missing values using a justified imputation method.
- Encode relevant categorical variables and prepare the predictor set.
- Split the dataset into training and test sets.
- Train a Logistic Regression classifier to estimate the probability of delinquency.
- Select an intervention threshold using business considerations and validate it on held-out data.
- Review performance, fairness, limitations, and whether predictions are appropriate for the intended use.

6.3 Evaluation Strategy

Precision indicates the share of customers predicted as delinquent who are truly delinquent; recall indicates the share of actual delinquent cases the model identifies; F1 balances precision and recall; AUC-ROC measures class separability across thresholds; and a confusion matrix shows true/false positives and negatives. When classes are imbalanced, accuracy should not be the only metric considered.

6.4 Why Logistic Regression

Logistic Regression is a suitable transparent baseline for a binary financial prediction task. It is relatively easy to interpret and can output a probability, which is useful when business teams need to choose thresholds rather than rely on an unexplained yes/no result.

7. Task 3 — Business Report and Data Storytelling

The simulation emphasizes turning an analytical insight into a recommendation that a non-technical stakeholder can understand and act upon. The strongest style is to state the risk pattern, identify the relevant segment, explain the business implication, and define a measurable action in plain language.

7.1 Example SMART Recommendation

A strong example from the knowledge checks is a six-week SMS outreach pilot targeting customers under 30 with 2+ missed payments, with a goal of reducing 30-day delinquency by 10%. The value of this format is that it is specific, measurable, actionable, relevant to collections, and time-bound.

For responsible deployment, any demographic targeting rule should be reviewed for fairness and business appropriateness before production use. The recommendation is therefore best treated as a simulation example, not a universal rule.

8. Task 4 — AI-Driven Collections Strategy

8.1 Role of Agentic AI

Agentic AI differs from traditional rule-based automation because it can make goal-oriented decisions, adapt to changing context, and learn from outcomes. In the collections scenario, low-impact routine actions can be automated while high-impact decisions can remain subject to human review.

8.2 Learning Loop

The learning loop continuously monitors outcomes and uses past performance to refine future actions. A practical cycle is: predict risk → choose an action → observe customer outcome → measure effectiveness and fairness → update the decision strategy.

8.3 Responsible AI Guardrails

- Fairness checks to detect systematically different outcomes across customer groups.
- Explainability so customers and business stakeholders can understand why a high-risk flag was generated.
- Human review for high-impact decisions, including sensitive or individualized hardship actions.
- Regulatory and privacy controls appropriate to financial-services use cases.
- Ongoing monitoring for model drift, changing customer behavior, and unintended outcomes.

9. Integrated Solution Architecture

The four tasks can be viewed as one connected workflow rather than isolated activities. The analytical pipeline begins with customer data and ends with a monitored, responsible collections action that feeds new evidence back into the learning loop.



Figure 1. Integrated analytics-to-action workflow

10. Tools, Skills and Professional Practices

SKILL AREA	PRACTISED CAPABILITY
Data & Analytics	Exploratory Data Analysis, data-quality assessment, risk profiling, data interpretation
Predictive Analytics	Model selection, Logistic Regression, threshold thinking, model validation
Business Communication	Data storytelling, plain-language recommendations, SMART action design
AI Strategy	Agentic AI, process automation, learning loops, decision support
Responsible AI	Fairness, explainability, human oversight, regulatory alignment, privacy
GenAI	Using GenAI to accelerate analytical reasoning, structure model plans, and develop business-facing outputs

11. Key Learning Outcomes

- Connected EDA, predictive modeling, and business decision making into one end-to-end analytics workflow.
- Learned to treat data quality and missing-value handling as modeling decisions rather than simple preprocessing.
- Practised communicating model findings in business language rather than relying only on technical metrics.
- Understood how agentic AI and learning loops can improve collections workflows while retaining human accountability.
- Recognized fairness, explainability, privacy, and regulatory alignment as core parts of financial AI design.

12. Conclusion and Future Scope

The Tata GenAI Powered Data Analytics Job Simulation provided an integrated view of how modern data and AI capabilities can support financial-services collections. The strongest lesson is that a useful AI system is more than a predictive model: it needs sound data exploration, understandable recommendations, responsible automation, measurable outcomes, and a feedback loop.

Future scope includes experimenting with additional classification models, calibrated probability thresholds, fairness monitoring across customer groups, controlled intervention experiments, and production-style monitoring of model drift and business outcomes.

13. References

- Forage, Tata Group — GenAI Powered Data Analytics Job Simulation (program page and task overview).
- Forage, Certificate of Completion issued to Abhinav Kumar Tripathi, 31 August 2026.
- Tata Data Analytics Glossary provided for the simulation tasks.
- Task materials and knowledge checks completed within the Tata Forage simulation.

Appendix A — Certificate of Completion

The source report contains a certificate of completion issued by Forage. The certificate image below is reproduced from the supplied reference PDF for documentation purposes. It retains the name shown on the original certificate; the academic report identity above has been updated to ABHINAV KUMAR TRIPATHI, Roll No. 2410031511.



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Abhinav Tripathi

Data Visualisation: Empowering Business with Effective Insights

Certificate of Completion
August 30th, 2026

Over the period of August 2026, Abhinav Tripathi has completed practical tasks in:

Framing the Business Scenario
Choosing the Right Visuals
Creating Effective Visuals
Communicating Insights and Analysis

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6a946e0f6282564772d12fe7 | User Verification Code 6a946db16ecab27e0ed06edd | Issued by Forage

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